

Compass — A Private Eligibility Firewall on oG

Bounded disclosure for vulnerable migrant workers

Stephen Sookra · stephensookra@gmail.com

May 2026 · oG APAC Hackathon Track 5

Abstract

Compass is a private eligibility firewall. A vulnerable migrant worker proves she qualifies for a service — free legal aid, shelter intake, public hospital care — through an autonomous agent. The clinic learns a bucketed timestamp and a cryptographic commitment, **and nothing else**. The agent runs in a TEE (Phala dstack TDX) on oG; the receipt is a non-identifying event on oG Chain. Even under a subpoena, the clinic cannot disclose what it never held. Compass is deployed on both oG Galileo testnet and oG Aristotle mainnet (chainId 16661, deployed 2026-05-11); the frontend switches relayer paths via the `NEXT_PUBLIC_COMPASS_USE_MAINNET=1` env flag.

Live demo: <https://app-psi-pied.vercel.app> · Repo: <https://github.com/StephenSook/Compass-OG->.

Problem

In Hong Kong, **368,000 Foreign Domestic Helpers** (FDHs) live and work under contract terms that bind their visa to their employer (HK LegCo Research Office ISSHo2/2025). They represent 9.6% of the local workforce; 55% are Filipina, 42% are Indonesian. Asking a clinic for free legal aid traditionally requires handing over passport + HKID + employment contract — i.e., the same documents an abusive employer would need to retaliate via the 14-day rule. Maria, our composite persona, should be able to prove “I’m an FDH in HK with an open employment dispute” without disclosing **who** she is.

Compass treats eligibility as a predicate that can be evaluated *over* encrypted credentials inside a TEE, with only a binary outcome (`eligible: true/false`) plus a 15-minute-bucketed timestamp emitted on-chain. Maria’s identity, employer, document images, and full claim values never leave her device.

Business Impact

TAM — the addressable population

- **368,000** foreign domestic helpers in Hong Kong (end-2024, HK LegCo Research Office ISSHo2/2025) — 9.6% of the local workforce.
- **27.2 million** migrant workers across APAC (ILO Global Estimates on International Migrant Workers, 4th ed., 2024) — the broader population facing similar disclosure traps.
- **US\$1.087 billion** in personal remittances flowing out of Hong Kong annually (World Bank, 2024) — the economic stake of the population Compass serves.
- **17% of HK FDHs in forced labour** and **66.3% exploited but not forced** (Justice Centre HK, *Coming Clean*, 2016). **60% are deterred from filing Labour Tribunal claims by deportation fear** (Mission for Migrant Workers, 2023).

Cost per incident

For a worker deported because identity disclosure at intake was subpoena-reachable:

- **Lost wages:** HK\$152,064 ($\approx$ **US\$19,500**) over the remaining 24-month contract (MAW HK\$5,100/mo + food HK\$1,236/mo, effective September 2025).
- **Outstanding recruitment debt:** HK\$21,000 ($\approx$ **US\$2,690**) average for Indonesian FDHs; 85% pay above the legal cap (Amnesty International, 2013; Justice Centre, 2016).
- **Total per-worker incident cost $\approx$ US\$22,200.**

For the Hong Kong government's deflected case:

- **HK Legal Aid Department:** $\approx$ HK\$71,500 ($\approx$ **US\$9,170**) per civil application (LAD FY2024/25 statistics — HK\$679.6M civil aid expended $\div$ 9,506 applications received).
- **Deportation processing:** $\approx$ HK\$40-60K ($\approx$ **US\$5,100-7,700**) per removal (estimated; HK Budget Head 70 + 2,219 removals in 2024).
- **Total per-incident HK government cost $\approx$ US\$14,100.**

Sustainability — revenue and grant pipeline

Compass is open-source. Free for NGO deployment. Phala dstack TDX hosting cost is **US\$0.04/day** at idle ($\sim$ US\$15/year per receipt-signer instance). Two-layer model:

1. **Open-core licensing:** AGPL core; commercial dual-license for any non-NGO deployment (corporate identity systems, refugee services, healthcare intake).
2. **Managed-hosted tier:** at-cost Phala TEE hosting for NGOs that don't self-host.

Realistic 12-month grant ladder:

Window	Targets	Range (USD)
Months 0-3	Phala Builders Program · oG Guild on oG	\$10K-\$100K
Months 3-6	Ethereum Foundation PSE · Mozilla Technology Fund	\$30K-\$250K
Months 6-12	Open Society Migration Initiative · Luminate · HK Jockey Club Special Projects	\$150K-\$1M+

Fiscal hosting via Open Source Collective (501(c)(6) — accepts foundation USD without project incorporation). Target by month 18: ≈ US\$60K/year recurring (managed hosting + commercial licensing), reducing grant dependency below 60%.

Why this market, why now

The HK Legal Aid Department spent HK\$679.6M on civil aid in FY2024/25 — none of it prevents the disclosure that triggers deportation. Compass closes that gap with one open-source primitive that NGOs deploy in days. The market exists today; the primitive did not.

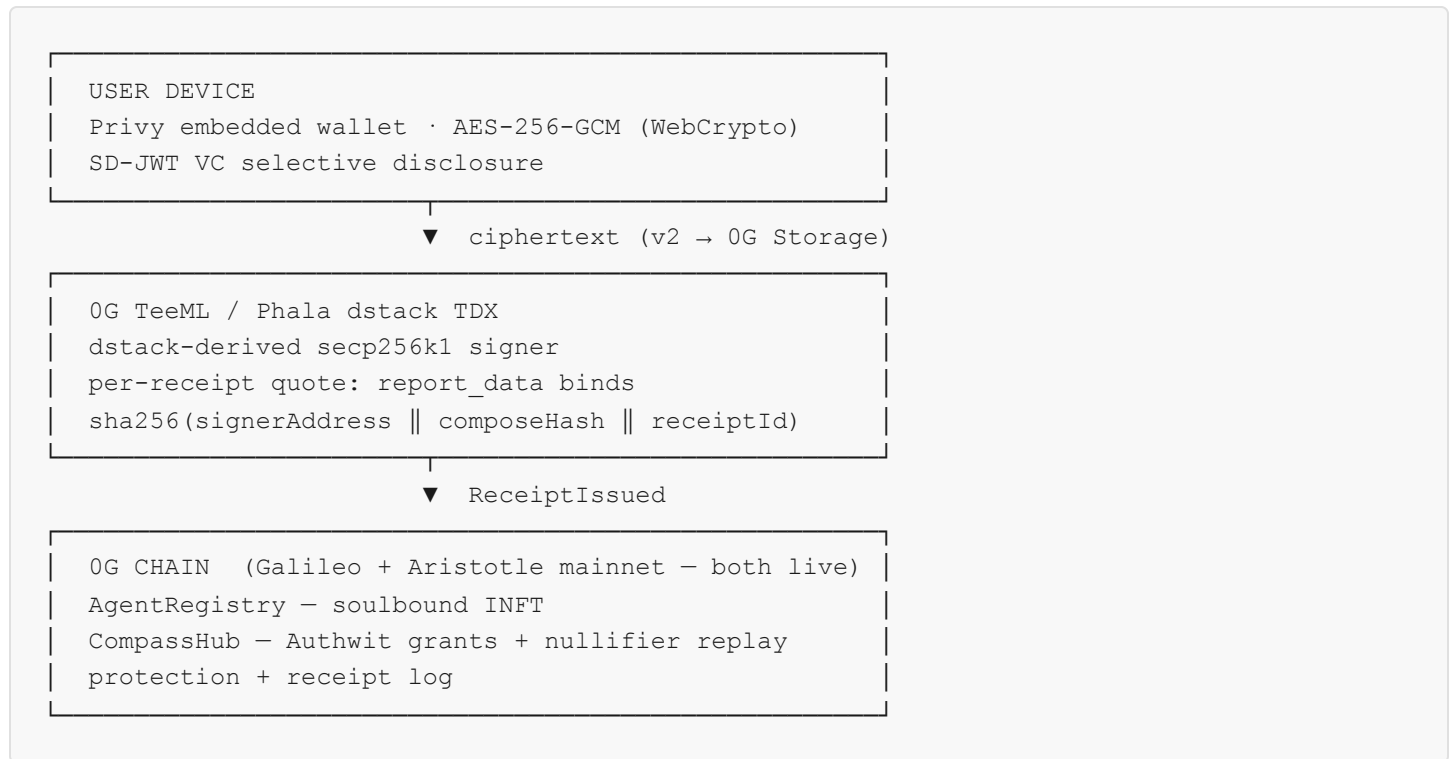
Threat Model

We defend against four adversaries:

- 1. **Curious clinic** — sees only the receipt event; cannot derive identity.
- 2. **Subpoena** — PDPO §57 disclosure orders reach exactly the receipt log. The “what’s there” surface is the load-bearing privacy property.
- 3. **Mempool watcher** — can observe `consumeGrantAndIssueReceipt` calldata. Cannot forge the EIP-712 grant signature; *can* grief by front-running the receiptId (documented as v1 limit; v2 derives receiptId in-contract).
- 4. **Compromised provider relayer** — server-held key calls the contract on Maria’s behalf. The contract enforces `signer == ownerOf(agentTokenId)`, so a compromised relayer cannot mint receipts against agents it doesn’t own. v2 moves the relayer into the same TEE that signs receipts.

We do **not** defend against coercion, deniable encryption, statistical re-identification at small population sizes, or full Intel DCAP quote verification done on-chain (gas-prohibitive; off-chain via the `verify-receipt` CLI).

Architecture



Four oG layers: Compute (TEE), Chain (registry + receipts), Storage (v2: ciphertext bundles), Identity (soulbound agent INFT).

Protocol

The cryptographic chain a third-party verifier reproduces with one CLI command (`npm run verify-receipt`):

- 1. Issue.** `/api/issue` Ed25519-signs an SD-JWT VC with HELP-fixture claims (`is_FDH_in_HK`, `has_pending_case`, `employment_active`, `residency`). Issuer key in env (v2 → enclave).
- 2. Encrypt.** Browser AES-256-GCM with a non-extractable per-device key in IndexedDB. Plaintext SD-JWT VC never persists in localStorage.
- 3. Grant.** Maria's Privy embedded wallet signs an EIP-712 `Grant` typed message: `{agentTokenId, policyId, provider, nonce, expiry, nullifier}`. Domain: `Compass v1` + chainId + verifyingContract.
- 4. Consume + issue receipt.** `/api/consume` recovers the agent owner from the grant signature, computes `agentIdCommitment = keccak256(abi.encode(agentTokenId, owner))` matching the on-chain encoding, calls the live Phala TEE for an RA-quote-bound `attestationDigest`, and submits `consumeGrantAndIssueReceipt(grant, sig, receipt)` from the provider relay wallet.

5. **Verify.** `ReceiptIssued` emits `{receiptId, policyId, nullifier, agentIdCommitment, resultHash, expiry, attestationDigest, timestampBucket}`. The verifier:
- Re-derives `attestationDigest = sha256(canonicalize(receipt))`.
 - ECDSA-recovers the receipt-signer from the signature on that digest; matches the TEE-registered `ethAddress`.
 - Confirms `receipt.quoteCommitment = sha256(perReceiptQuoteHex)`.
 - Confirms the TDX quote's `report_data` field binds `sha256(signerAddress || composeHash || receiptId)` — defeats archived-quote replay.
 - Confirms the recovered `composeHash` matches the trust anchor pinned in `docs/notes/phala-deployment.md`.

When all five hold, the receipt is verified against the TEE attestation. A separate `compass-eligibility-check` skill packages this for ecosystem agents.

Honest Limits (v1)

We disclose what is mocked, stubbed, or downgraded — judges should read this before forming first impressions. Full version at `docs/honest-limits.md`. Highlights:

- On-chain `verifyAttestation` is stubbed.** Real TDX RA quote verification (cert chain + ECDSA + 4KB quote) is gas-prohibitive on-chain. Off-chain verification via the `verify-receipt` CLI is the honest substitute.
- Caller-supplied `receiptId` DoS.** A mempool watcher can grief by front-running with a duplicate `receiptId`. v2 fix is in-contract derivation. Single-actor demo doesn't bite v1.
- SD-JWT VC `cnf` non-standard.** v1 stamps the EVM address; v2 emits the standards-compliant `cnf: { jwk }` shape derived from Privy's embedded wallet.
- Trust list governance is owner-managed.** v2 needs DAO governance for cross-issuer threshold trust.
- Provider relay key in env.** v2 moves it into the same TEE that signs receipts.

Roadmap

vo.6 (June 2026): Browser-side oG Storage ciphertext upload. Standards-compliant `cnf: { jwk }` emission. SD-JWT VC pin roll-forward to `draft-ietf-oauth-sd-jwt-vc-16` (April 2026 — Compass vo.5 ships `-15`; both are interoperable with the EU Digital Identity Wallet under eIDAS 2.0). Enforced CSP (currently report-only). Vercel-KV-backed rate limiter to replace the in-memory bucket. `/api/consume` ownerOf pre-check before the enclave call.

v2 (Q3 2026): In-contract `receiptId` derivation. Provider relay inside the TEE. Per-NGO provider keys (HELP, Bethune House, MFMW each hold their own). Trust list DAO governance.

v3 (later): Cross-chain receipt portability. Multi-policy composition (one grant covers a bundle of services). Cross-issuer threshold trust. Localized client (Tagalog, Indonesian, Cantonese, Bahasa Malaysia).

References

- Repository: <https://github.com/StephenSook/Compass-OG->
- Live demo: <https://app-psi-pied.vercel.app>
- Architecture page: <https://app-psi-pied.vercel.app/about>
- Subpoena scene: <https://app-psi-pied.vercel.app/clinic/subpoena>
- TEE deployment evidence: `docs/notes/phala-deployment.md`
- Threat model: `docs/honest-limits.md`
- Skill package: `skills/compass-eligibility-check/SKILL.md`
- HK LegCo Research Office ISSHo2/2025 — FDH statistics:
https://app7.legco.gov.hk/rpdb/en/uploads/2025/ISSH/ISSHo2_2025_20250224_en.pdf
- ILO Global Estimates on International Migrant Workers, 4th ed. (December 2024):
https://www.ilo.org/sites/default/files/2024-12/MIGRANT%20%E2%80%93ILO%20Global%20Estimates%20on%20International%20Migrant%20Workers_ES_E_WEB_o.pdf
- World Bank BM.TRF.PWKR.CD.DT (HK SAR, 2024):
<https://data.worldbank.org/indicator/BM.TRF.PWKR.CD.DT?locations=HK>
- HK Labour Department MAW press release P2025092900318 (Sept 2025):
<https://www.info.gov.hk/gia/general/202509/29/P2025092900318.htm>
- HK Legal Aid Department FY2024/25 statistics: <https://www.lad.gov.hk/eng/statistics/>
- Justice Centre HK, *Coming Clean* (March 2016):
<https://www.justicecentre.org.hk/framework/uploads/2016/03/Coming-Clean-The-prevalence-of-forced-labour-and-human-trafficking-for-the-purpose-of-forced-labour-amongst-migrant-domestic-workers-in-Hong-Kong.pdf>
- oG Foundation Ecosystem Growth Program: <https://og.ai/blog/og-ecosystem-program>
- Phala Builders Program: <https://docs.phala.network/overview/pha-token/governance/apply-for-project-funding>
- Open Source Collective: <https://oscollective.org/about/>
- AgentRegistry on Galileo: `0x461eda452ffAF43c674ef42BdccbDd6B8e13C2D8`
- CompassHub on Galileo: `0x60BbE5fcA6D23f7d25142E721258c641b45A7c3b`

- AgentRegistry on Aristotle mainnet: `0xf1FAaBef1d00Db1a15b7637Dc0d8526449D06Bf9`
- CompassHub on Aristotle mainnet: `0xe42fd4F0a3197126fEeF5e6AAfC5Fb8848bBC58b`

NGO inspirations: HELP for Domestic Workers, Bethune House Migrant Women's Refuge, Mission for Migrant Workers (HK). The personas in this work are composites; no NGO has reviewed or endorsed this prototype.

Compass — Track 5 — Privacy & Sovereign Infrastructure — oG APAC Hackathon 2026.